

DataPrepAI

Dataset Analysis Report

Dataset: /content/Diabetes Missing Data.csv

1. Dataset Overview

Metric	Value
Dataset Name	Diabetes Missing Data.csv
Rows	768
Columns	9
Memory Usage	0.05 MB
Numerical Columns	9
Categorical Columns	0

Dataset Preview

Pregnant	Glucose	Diastolic_BP	Skin_Fold	Serum_Insulin	BMI	Diabetes_Pedigree	Age	Class
6.0	148.0	72.0	35.0	nan	33.6	0.627	50.0	1.0
1.0	85.0	66.0	29.0	nan	26.6	0.351	31.0	0.0
8.0	183.0	64.0	nan	nan	23.3	0.672	32.0	1.0
1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0

2. Data Quality Analysis

Missing Value Analysis

Column	Missing Values	Missing Percentage (%)	Status
Pregnant	0	0.0	Clean
Glucose	5	0.65	Missing
Diastolic_BP	35	4.56	Missing
Skin_Fold	227	29.56	Missing
Serum_Insulin	374	48.7	Missing
BMI	11	1.43	Missing
Diabetes_Pedigree	0	0.0	Clean
Age	0	0.0	Clean
Class	0	0.0	Clean

Total Missing Values: 652
Columns with Missing Values: 5
Dataset Missing Percentage: 9.43%

Duplicate Analysis

Metric	Value
Total Rows	768.0
Duplicate Rows	0.0
Non-Duplicate Rows	768.0
Duplicate Percentage (%)	0.0

Outlier Analysis

Column	Outlier Count	Status
Pregnant	4	Outliers Detected
Glucose	0	Clean
Diastolic_BP	14	Outliers Detected
Skin_Fold	3	Outliers Detected
Serum_Insulin	24	Outliers Detected
BMI	8	Outliers Detected
Diabetes_Pedigree	29	Outliers Detected
Age	9	Outliers Detected
Class	0	Clean

Total Outliers: 91
Columns with Outliers: 7
Clean Columns: 2

Constant Column Analysis

No constant columns detected.

Missing Value Recommendations

Column	Missing Values	Recommended Technique	Reason
Glucose	5	Mean Imputation	No outliers detected
Diastolic_BP	35	Median Imputation	Outliers detected
Skin_Fold	227	Median Imputation	Outliers detected
Serum_Insulin	374	Median Imputation	Outliers detected
BMI	11	Median Imputation	Outliers detected

3. Statistical Analysis

Numerical Statistical Analysis

Column	Count	Mean	Median	Standard Deviation	Minimum	Q1	Q3	Maximum	Range
Pregnant	768	3.85	3.0	3.37	0.0	1.0	6.0	17.0	17.0
Glucose	763	121.69	117.0	30.54	44.0	99.0	141.0	199.0	155.0
Diastolic_BP	733	72.41	72.0	12.38	24.0	64.0	80.0	122.0	98.0
Skin_Fold	541	29.15	29.0	10.48	7.0	22.0	36.0	99.0	92.0
Serum_Insulin	394	155.55	125.0	118.78	14.0	76.25	190.0	846.0	832.0
BMI	757	32.46	32.3	6.92	18.2	27.5	36.6	67.1	48.9
Diabetes_Pedigree	768	0.47	0.37	0.33	0.08	0.24	0.63	2.42	2.34
Age	768	33.24	29.0	11.76	21.0	24.0	41.0	81.0	60.0
Class	768	0.35	0.0	0.48	0.0	0.0	1.0	1.0	1.0

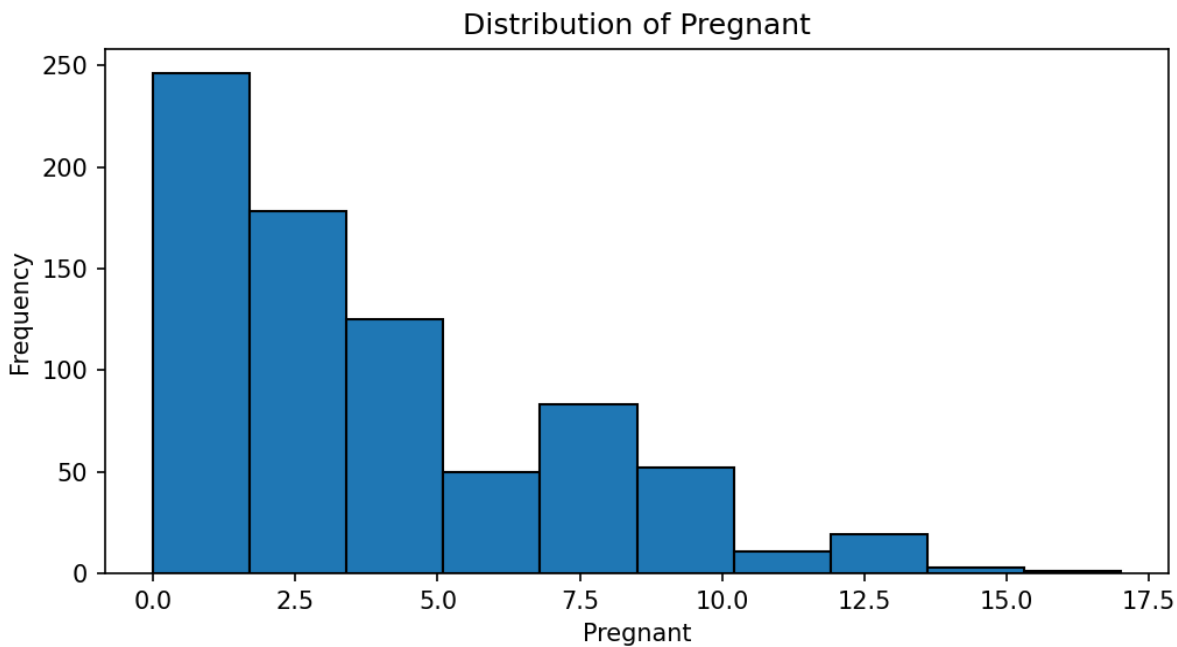
Categorical Statistical Analysis

No categorical columns detected.

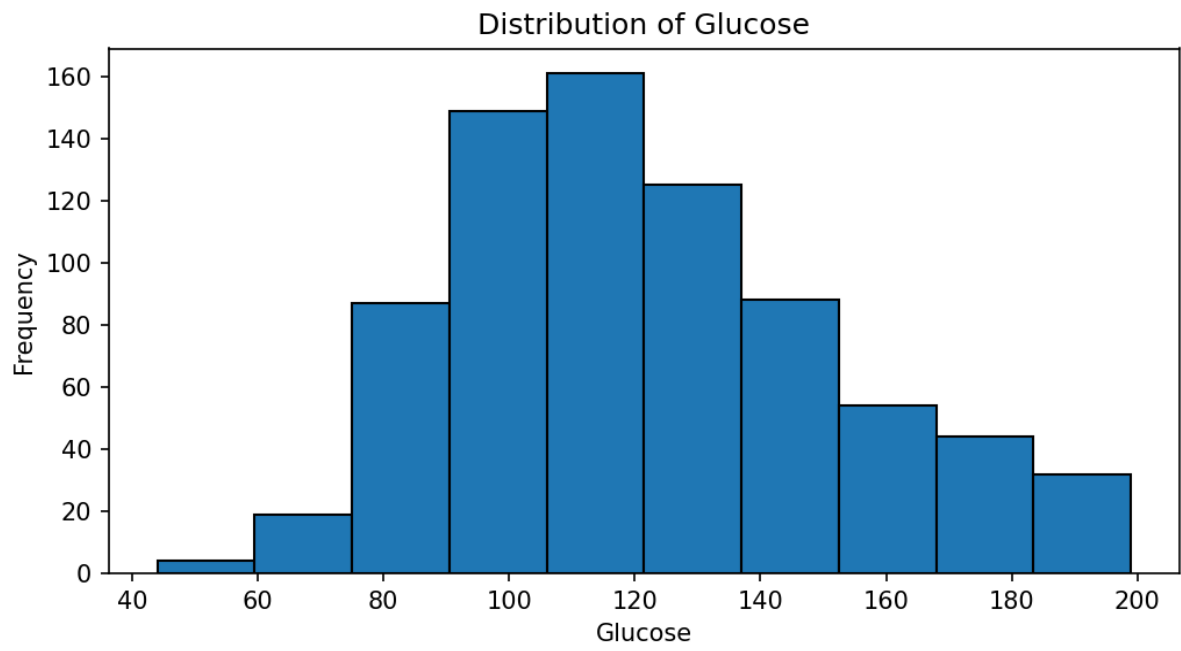
4. Distribution Analysis

Numerical Distributions

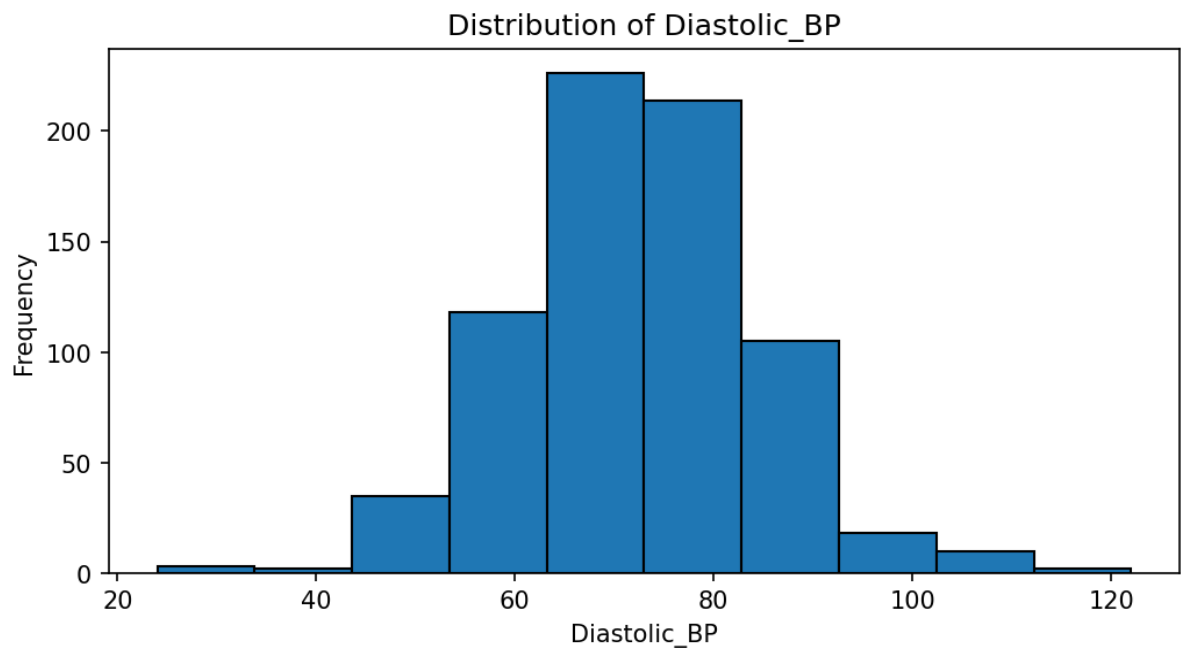
Pregnant



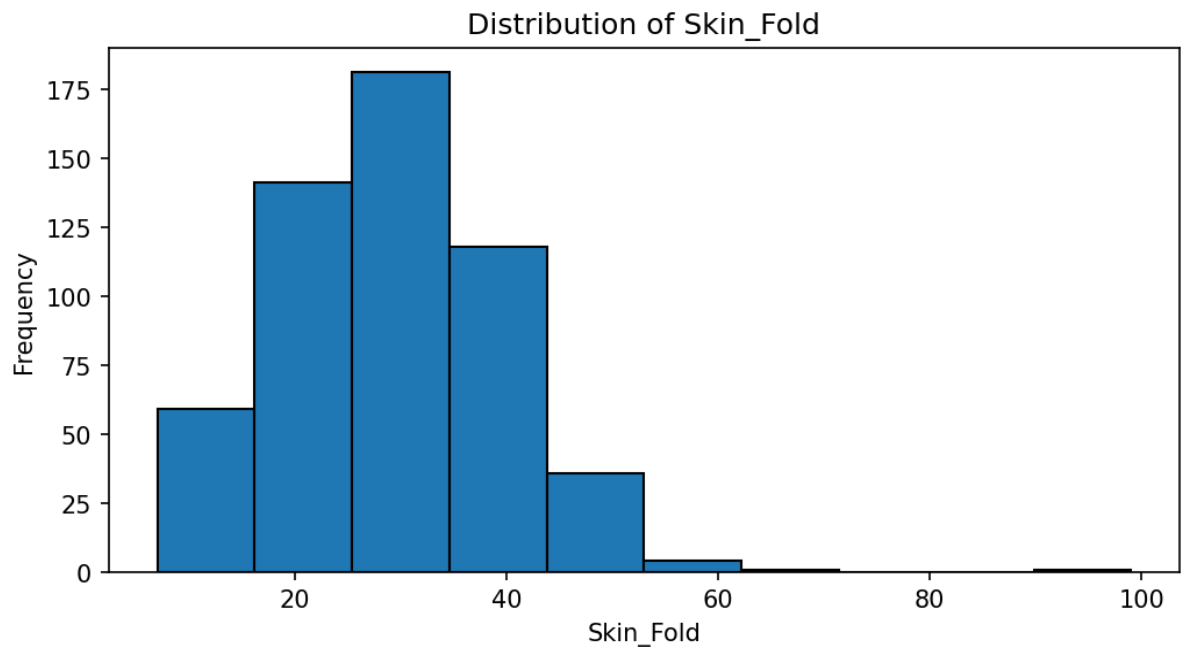
Glucose



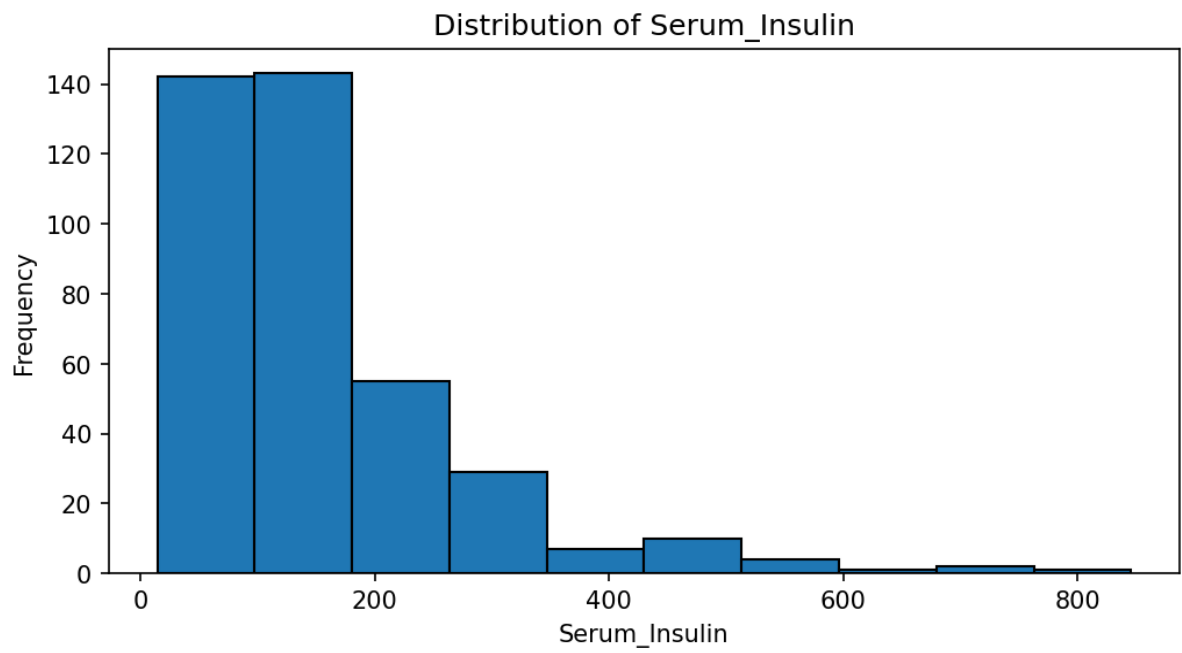
Diastolic_BP



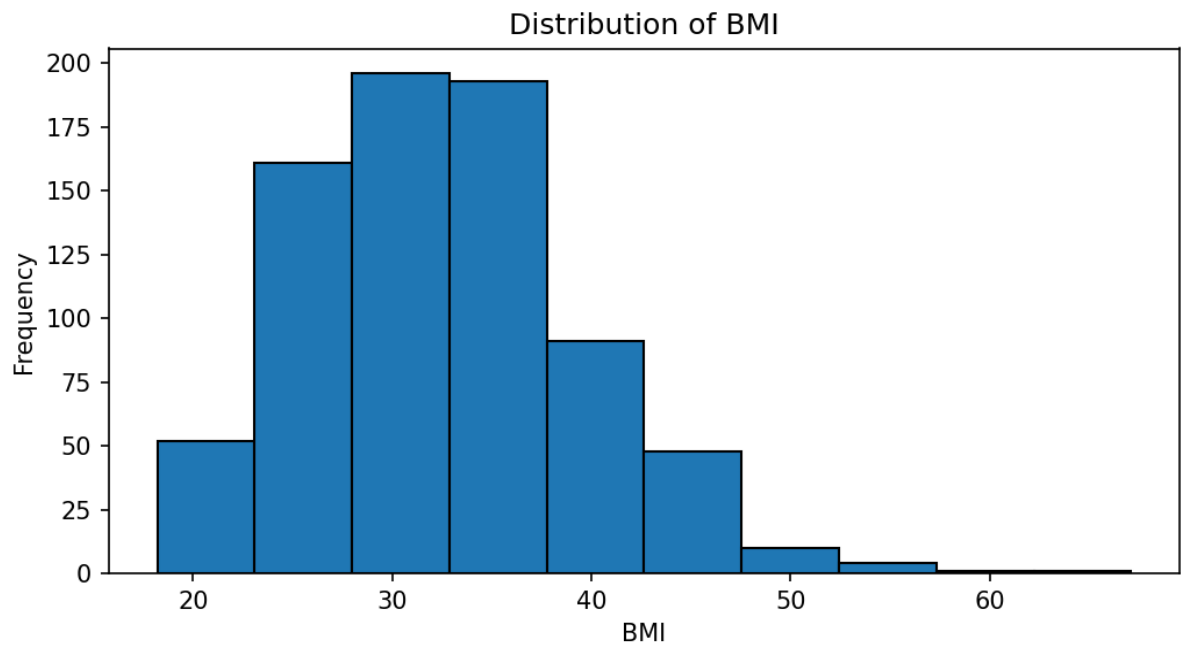
Skin_Fold



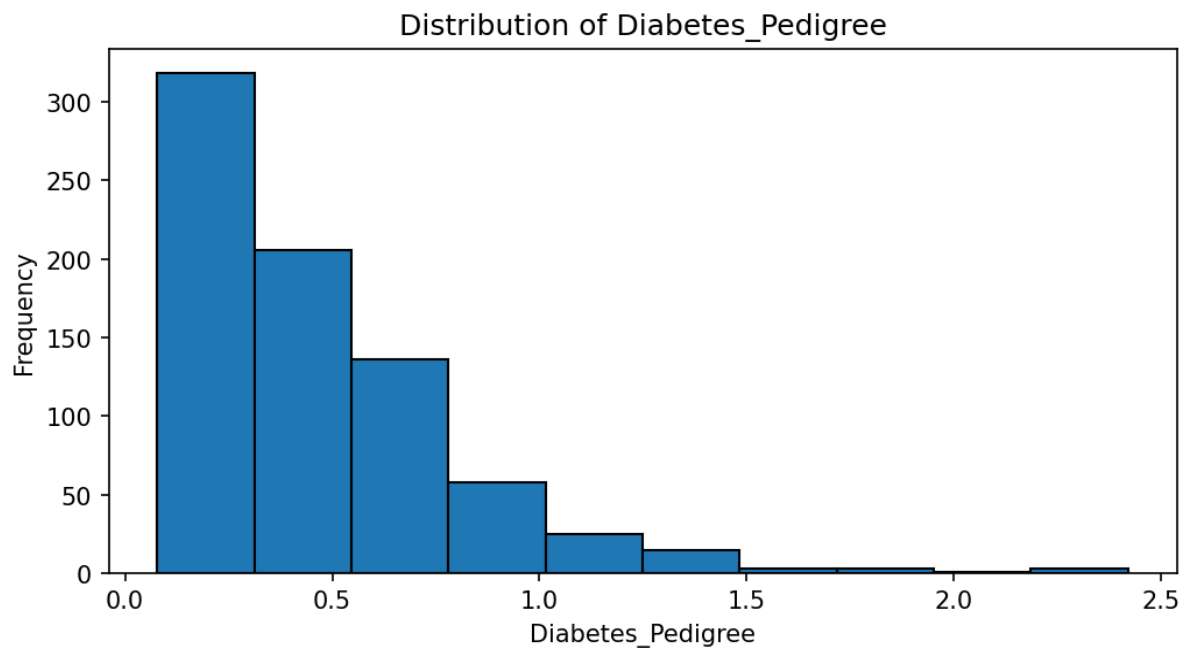
Serum_Insulin



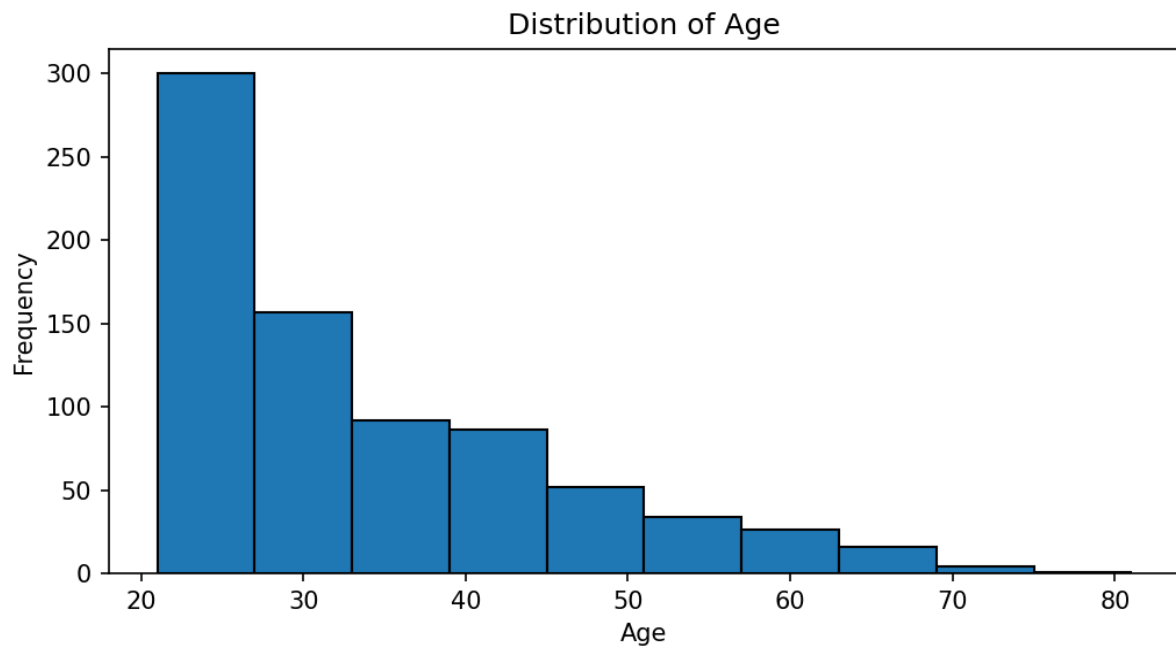
BMI



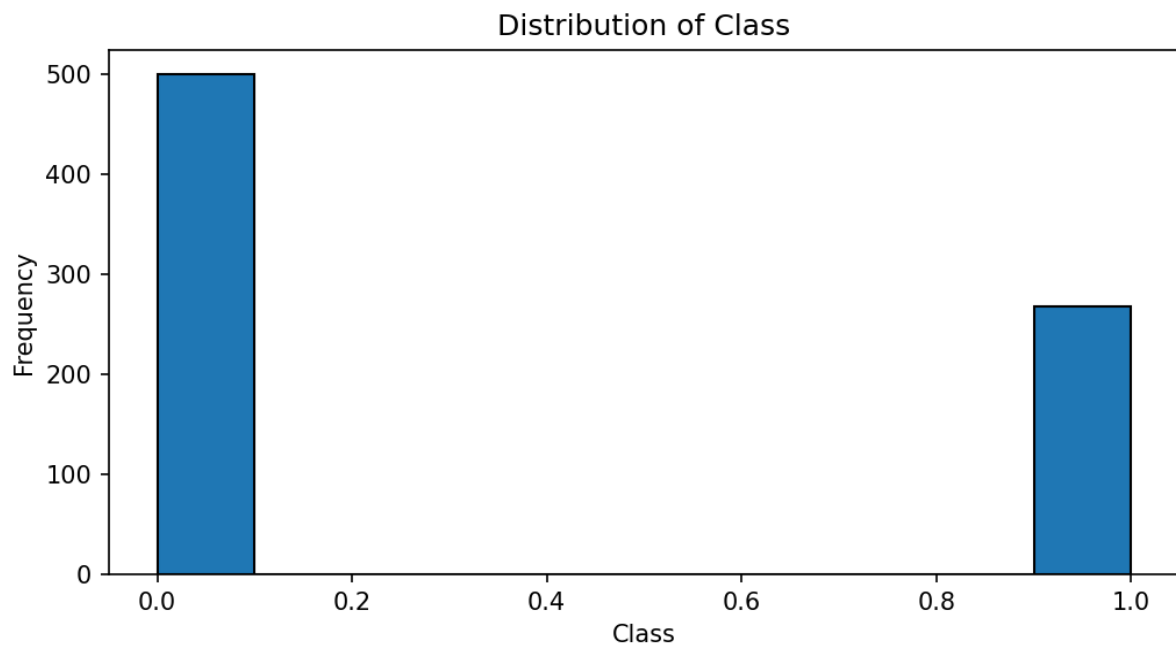
Diabetes_Pedigree



Age



Class



5. Relationship / Correlation Analysis

Correlation Matrix

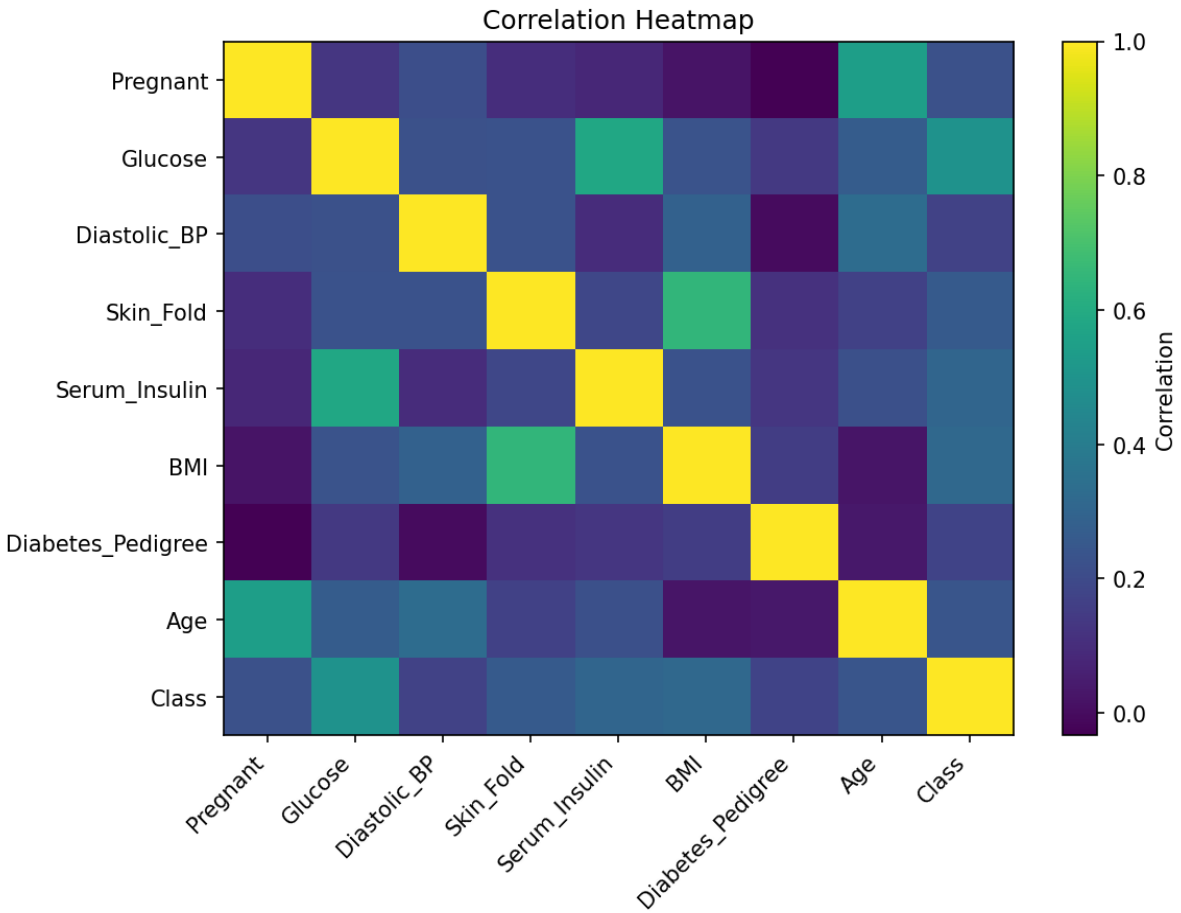
	Pregnant	Glucose	Diastolic_BP	Skin_Fold	Serum_Insulin	BMI	Diabetes_Pedigree	Age	Class
Pregnant	1.0	0.13	0.21	0.1	0.08	0.02	-0.03	0.54	0.22
Glucose	0.13	1.0	0.22	0.23	0.58	0.23	0.14	0.27	0.49
Diastolic_BP	0.21	0.22	1.0	0.23	0.1	0.29	-0.0	0.33	0.17
Skin_Fold	0.1	0.23	0.23	1.0	0.18	0.65	0.12	0.17	0.26
Serum_Insulin	0.08	0.58	0.1	0.18	1.0	0.23	0.13	0.22	0.3

	Pregnant	Glucose	Diastolic_BP	Skin_Fold	Serum_Insulin	BMI	Diabetes_Pedigree	Age	Class
BMI	0.02	0.23	0.29	0.65	0.23	1.0	0.16	0.03	0.31
Diabetes_Pedigree	-0.03	0.14	-0.0	0.12	0.13	0.16	1.0	0.03	0.17
Age	0.54	0.27	0.33	0.17	0.22	0.03	0.03	1.0	0.24
Class	0.22	0.49	0.17	0.26	0.3	0.31	0.17	0.24	1.0

Strongest Relationships

Relationship	Columns	Correlation
Strongest Positive	Skin_Fold ↔ BMI	0.65
Strongest Negative	Pregnant ↔ Diabetes_Pedigree	-0.03
Weakest	Diastolic_BP ↔ Diabetes_Pedigree	-0.00

Correlation Heatmap



End of DataPrepAI Report